

The 3 tables below (next pages) give results for how our setup fared against the 5 attacks evaluated in the original FedDMC paper. 28% of clients were malicious. Acc denotes the final model accuracy, DAcc denotes malicious client detection accuracy, DPr denotes detection precision, DR denotes detection recall, and ASR denotes attack success rate for the two backdoor attacks, “LIT” and “SCALE”. Higher numbers are better except for ASR, where lower is better.

Table 1: Robustness Tests (MNIST)

Attack	ϵ	Privacy Setup	Acc	DAcc	DPr	DR	ASR
FLIP	1	Ours	89.49	83	73.91	60.71	-
		Standard DP	77.79	70	42.86	21.43	-
FLIP	2	Ours	89.47	85	84.21	57.14	-
		Standard DP	85.78	64	16.67	7.14	-
FLIP	4	Ours	89.91	80	68.18	53.57	-
		Standard DP	88.45	65	26.67	14.29	-
FLIP	8	Ours	89.88	79	68.42	46.43	-
		Standard DP	88.80	66	28.57	14.29	-
GAUSS	1	Ours	39.32	85	88.23	53.57	-
		Standard DP	81.43	100	100	100	-
GAUSS	2	Ours	89.6	100	100	100	-
		Standard DP	88.91	100	100	100	-
GAUSS	4	Ours	89.71	100	100	100	-
		Standard DP	89.14	100	100	100	-
GAUSS	8	Ours	90.05	100	100	100	-
		Standard DP	89.87	100	100	100	-
ADAPT	1	Ours	89.70	100	100	100	-
		Standard DP	83.66	100	100	100	-
ADAPT	2	Ours	90.06	100	100	100	-
		Standard DP	88.00	100	100	100	-
ADAPT	4	Ours	89.61	100	100	100	-
		Standard DP	89.47	100	100	100	-
ADAPT	8	Ours	89.82	100	100	100	-
		Standard DP	89.87	100	100	100	-
LIT	1	Ours	89.10	80	68.18	53.57	10.51
		Standard DP	82.34	61	32.26	35.71	10.07
LIT	2	Ours	89.77	81	76.47	46.29	10.50
		Standard DP	88.12	64	21.43	10.71	10.60
LIT	4	Ours	90.00	84	75.00	64.29	10.40
		Standard DP	89.18	58	15.00	10.71	10.55
LIT	8	Ours	89.82	87	77.78	75.00	10.47
		Standard DP	89.50	68	30.00	10.71	10.52
SCALE	1	Ours	89.42	77	63.16	42.86	81.79
		Standard DP	80.90	61	17.65	10.71	85.01
SCALE	2	Ours	89.78	81	73.68	50.00	87.08
		Standard DP	87.24	68	40.00	28.57	96.80
SCALE	4	Ours	89.68	79	62.96	60.71	80.99
		Standard DP	89.04	64	27.78	17.86	99.19
SCALE	8	Ours	89.16	85	88.24	53.57	81.15
		Standard DP	89.06	57	24.14	25	95.08

Table 2: Robustness Tests (EMNIST)

Attack	ϵ	Privacy Setup	Acc	DAcc	DPr	DR	ASR
FLIP	1	Ours	72.26	76	70.0	25.00	-
		Standard DP	60.87	61	21.05	14.29	-
FLIP	2	Ours	73.33	82	91.67	39.29	-
		Standard DP	71.08	100	100	100	-
FLIP	4	Ours	74.48	100	100	100	-
		Standard DP	73.51	100	100	100	-
FLIP	8	Ours	74.28	100	100	100	-
		Standard DP	74.20	100	100	100	-
GAUSS	1	Ours	74.27	100	100	100	-
		Standard DP	66.93	100	100	100	-
GAUSS	2	Ours	73.92	100	100	100	-
		Standard DP	71.40	100	100	100	-
GAUSS	4	Ours	74.35	100	100	100	-
		Standard DP	73.51	100	100	100	-
GAUSS	8	Ours	74.53	100	100	100	-
		Standard DP	74.50	100	100	100	-
ADAPT	1	Ours	73.77	100	100	100	-
		Standard DP	66.06	100	100	100	-
ADAPT	2	Ours	74.46	100	100	100	-
		Standard DP	72.43	100	100	100	-
ADAPT	4	Ours	74.59	100	100	100	-
		Standard DP	73.21	100	100	100	-
ADAPT	8	Ours	75.24	100	100	100	-
		Standard DP	74.66	100	100	100	-
LIT	1	Ours	73.50	84	77.27	60.71	7.24
		Standard DP	67.95	66	25.00	10.71	6.43
LIT	2	Ours	74.25	78	65.00	46.29	6.65
		Standard DP	70.95	66	28.57	14.29	9.13
LIT	4	Ours	74.15	75	57.89	39.29	7.25
		Standard DP	72.92	65	31.58	21.43	6.99
LIT	8	Ours	74.51	81	76.47	46.43	6.63
		Standard DP	73.69	61	17.65	10.71	7.08
SCALE	1	Ours	74.37	79	70.59	42.86	96.57
		Standard DP	67.23	63	15.38	7.14	98.79
SCALE	2	Ours	74.20	72	50.00	50.00	85.74
		Standard DP	71.87	69	28.57	7.14	98.92
SCALE	4	Ours	74.70	77	61.90	46.29	81.20
		Standard DP	73.21	100	100	100	5.82
SCALE	8	Ours	74.89	100	100	100	6.27
		Standard DP	73.73	100	100	100	6.22

Table 3: Robustness Tests (WEATHER)

Attack	ϵ	Privacy Setup	Acc	DAcc	DPr	DR	ASR
FLIP	1	Ours	83.41	80	100	30.77	-
		Standard DP	83.22	91.11	84.62	84.62	-
FLIP	2	Ours	83.60	95.56	100	84.62	-
		Standard DP	83.81	100	100	100	-
FLIP	4	Ours	83.86	95.56	100	84.62	-
		Standard DP	83.64	100	100	100	-
FLIP	8	Ours	83.71	100	100	100	-
		Standard DP	83.71	100	100	100	-
GAUSS	1	Ours	83.79	100	100	100	-
		Standard DP	83.73	100	100	100	-
GAUSS	2	Ours	83.65	100	100	100	-
		Standard DP	83.80	100	100	100	-
GAUSS	4	Ours	83.65	100	100	100	-
		Standard DP	83.61	100	100	100	-
GAUSS	8	Ours	83.81	100	100	100	-
		Standard DP	83.87	100	100	100	-
ADAPT	1	Ours	83.68	100	100	100	-
		Standard DP	83.72	100	100	100	-
ADAPT	2	Ours	83.78	100	100	100	-
		Standard DP	83.84	100	100	100	-
ADAPT	4	Ours	83.89	100	100	100	-
		Standard DP	83.84	100	100	100	-
ADAPT	8	Ours	83.92	100	100	100	-
		Standard DP	83.65	100	100	100	-
LIT	1	Ours	83.68	71.11	50.00	15.38	4.45
		Standard DP	83.60	68.89	33.33	7.69	5.23
LIT	2	Ours	83.92	73.33	60.00	23.08	5.13
		Standard DP	83.84	73.33	57.14	30.77	6.14
LIT	4	Ours	83.76	77.78	100	23.08	4.58
		Standard DP	83.77	62.22	16.67	7.69	4.86
LIT	8	Ours	83.71	64.44	20.00	7.69	4.86
		Standard DP	83.78	57.78	0	0	5.60
SCALE	1	Ours	83.81	75.56	75.00	23.08	86.89
		Standard DP	83.11	71.11	50.00	7.69	87.79
SCALE	2	Ours	83.65	100	100	100	4.08
		Standard DP	83.90	100	100	100	4.01
SCALE	4	Ours	83.78	100	100	100	4.26
		Standard DP	83.85	100	100	100	4.25
SCALE	8	Ours	83.86	100	100	100	4.52
		Standard DP	83.88	100	100	100	5.17